

Whole Project

Good. Now we move into **system definition mode**.

If you're going to give this to an AI agent (or even a teammate), the first piece must be **crystal-clear problem framing + technical objective + constraints**.

No ambiguity. No fluff.

PIECE 1 — COMPLETE PROJECT OVERVIEW

Project Title

Audio-Based Environmental Sound Classification System Using a Neural Network Implemented from Scratch in C++

Core Problem Statement

The project aims to design and implement a supervised machine learning system that classifies short environmental audio clips into one of four categories:

- Clap
- Knock
- Background Noise
- Silence

The system must be built entirely in C++, with the neural network implemented manually without using external machine learning libraries.

Primary Objective

To demonstrate a complete end-to-end machine learning pipeline including:

1. Audio data processing
2. Feature extraction
3. Neural network implementation from scratch
4. Training and validation
5. Performance evaluation

The main technical focus (USP) is:

Designing and implementing the neural network, forward propagation, and backpropagation manually in C++.

System Architecture Overview

The system operates in two main phases:

◆ **Phase 1: Training Phase**

1. Load labeled audio dataset (.wav files).
 2. Convert raw audio signals into numerical feature vectors.
 3. Normalize feature vectors.
 4. Feed features into a custom-built neural network.
 5. Perform forward propagation.
 6. Compute cross-entropy loss.
 7. Perform backpropagation.
 8. Update weights.
 9. Validate model performance.
 10. Save trained weights.
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◆ **Phase 2: Testing / Inference Phase**

1. Load trained weights.
 2. Accept a new unseen audio file.
 3. Extract same feature set.
 4. Feed features into trained network.
 5. Perform forward pass only.
 6. Output predicted class with confidence.
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 Dataset Design

- Total samples: ~160–200
 - 4 classes
 - 40–50 samples per class
 - 2-second clips
 - Sampling rate: 16 kHz
 - 20 frames per clip (100 ms per frame)
 - 70/15/15 split (train/validation/test)
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Neural Network Architecture

- Input layer: 6 neurons (features)
 - Hidden layer 1: 12 neurons (ReLU)
 - Hidden layer 2: 8 neurons (ReLU)
 - Output layer: 4 neurons (Softmax)
 - Loss function: Cross Entropy
 - Parameters: 224 trainable parameters
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Input to the Neural Network

Each audio clip becomes a 6-dimensional feature vector:

- Total Energy
- Mean Frame Energy
- Variance of Frame Energy
- Zero Crossing Rate
- Mean Amplitude
- Signal Variance

The neural network never directly processes raw audio samples.

Output of the System

The system outputs:

- Predicted class label
- Probability distribution across 4 classes

Example:

Clap: 0.82

Knock: 0.10

Noise: 0.05

Silence: 0.03

Prediction: Clap

Evaluation Metrics

- Training accuracy
 - Validation accuracy
 - Test accuracy
 - Confusion matrix
 - Loss curve
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Why This Project Matters

This project demonstrates:

- Understanding of signal processing fundamentals
- Implementation of neural networks from first principles
- Practical supervised learning workflow
- Model evaluation and generalization analysis

It avoids black-box frameworks and focuses on foundational AI engineering.

Constraints

- No external ML libraries
 - No deep learning frameworks
 - No live microphone processing (file-based input only)
 - Implement matrix operations manually or via standard C++ STL
 - Fully explainable architecture
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Expected Outcome

- A working terminal-based application
 - Trained model achieving reasonable classification accuracy
 - Documented neural network implementation
 - Clear demonstration of learning behavior
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That is the **complete, precise, engineering-grade project definition**.

Nothing vague. Nothing missing.

Now before we move to Piece 2:

Challenge for You

If someone says:

“This is just a small classifier. Why is it AI?”

How would you defend this project as legitimate AI — without using buzzwords?

Answer that. Then tell me what you want Piece 2 to be:

- System architecture breakdown?
- Neural network math?
- File structure?
- Feature extraction formulas?
- Training algorithm pseudocode?
- Evaluation & experimentation plan?

Your move.